

DETECTING TAMPERED IMAGE BLOCKS USING ERROR CORRECTING CODE

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ABSTRACT

This paper presents an image authentication watermarking which is not only able to authenticate an entire image, but also detect the tampered image blocks. To this end, it encodes the image index using Error Correcting Code (ECC), and then amortizes the codeword symbols into the image blocks. This method reduces the size of block watermarks and hence provides good localizing accuracy.

1. INTRODUCTION

In the past few years, many image watermarking schemes (e.g., [1]) have been introduced to authenticate entire images. Nevertheless it is preferable to identify the tampered regions so as to use the intact regions if only one portion of an image is corrupted due to incidental data loss, or it is more valuable to find the temptation of the attacker from the tampered blocks. A straightforward localizing method [2] is to divide an image into blocks and embed integrity data into the blocks. However, this method is vulnerable to collage attack which assembles blocks of several authentic images or swaps blocks of the same image to forge a new authentic image [3]. To defeat the collage attack, a naïve method is to embed a unique image index, block number and block MAC¹ into the block. Consequentially, the challenge of localizing tampered regions is how to foil collage attack with smallest size of block watermarks, or finest localization granularity.

Wong and Memon [5] proposed to set image index to be the hash of the image so as to manage image indexes easily. However, this scheme completely impairs the localization ability because manipulation on a single pixel will alter the calculated image index, which results in different authentication watermark for all the blocks. Alternatively, the block overlapping scheme [6] calculates the block MACs from slightly larger blocks, but embeds the MACs into the original smaller blocks. The overlapping scheme [6] will localize the tampered positions with low accuracy.

¹Block hash: hash value of all the Most Significant Bit (MSB) planes, image index, and block number. A key-ed block hash is a block Message Authentication Code (MAC) [4].

Byun et al [7] modified singular values (SVs) of an image matrix to authenticate the image. The SVs and authentication logo are tiled and inserted into the Least Significant Bits (LSBs) of the original image. Regretfully, the authors made a mistake in extending the algorithm to localize tampered blocks because manipulation on one block may change all the SVs, and tiling operation will further propagate the SV error to all the block watermarks. As a result, it is not always possible to localize the tampered regions.

Fridrich et al [8] proposed an authentication scheme with a block logo shown in Figure 1. First, the originator divides an image into 8×16 blocks, calculates the block hash, and XORs the hash value with the logo to generate a token. Afterwards, the originator encrypts the token to generate the block watermark which is used to replace the LSBs of the block. To verify a block, the block watermark is extracted and decrypted into a token. The token is XORed with the block hash so as to recover the logo. If the first 52 bits of the recovered logo are equal to its next 52 bits, the block is authentic, otherwise, the block is tampered.

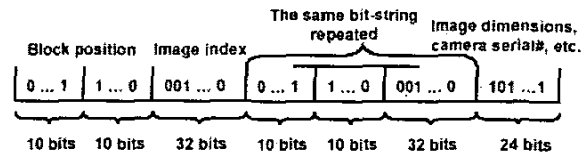


Fig. 1. Data structure of a logo for a specific block in [8].

Celik et al [9] [10] proposed a tree structure instead of image index to authenticate images. Each leaf node value is the MAC of an image block, while each non-leaf node value is the MAC of 4 child node values. Because the MAC of any non-leaf node is amortized into their lowest descendants (leaves or blocks), Celik's scheme requires $\lceil l/6 \rceil$ bits overhead for defeating collage attack, where l is the size of hash value. For example, If MD5 is used to generate MAC, the overhead is of at least $\lceil 128/6 \rceil = 22$ bits. In the aforementioned naïve scheme, the number of images labelled with 22-bit overhead is up to $2^{22} \approx 10^7$ which is

sufficiently large for most applications. Furthermore, FIPS PUB 180-2 [11] defines the new standard where $n \geq 256$. Therefore, the naïve method usually has advantages over the scheme [9] in terms of computational cost, localization accuracy, and robustness of tampering detection. Additionally, Celik's scheme can not decide whether separate authentic blocks belong to the same image or not, i.e., it can not merge separate blocks into one image.

The present authentication scheme employs image indexes to defeat collage attack too. But it encodes the image index with Error Correcting Code [12] instead of embedding the index directly. Then, a block watermark, including the block MAC, the block number and a codeword symbol, is embedded into LSBs of the block. Therefore, the present scheme can not only tolerate the legal cropping and malicious tampering, but also provide good localization accuracy. The experiments demonstrate the viability of detecting malicious tampering and region cropping.

The reminder of this paper is organized as follows. Section 2 describes the present image authentication scheme. Section 3 analyzes the performance in terms of security and localizing accuracy. The Experiment result is addressed in Section 4. Conclusion is drawn in Section 5.

2. LOCALIZING TAMPERED REGION SCHEME

This section describes the scheme for tampering detection and localization by embedding the Error Correcting Code (ECC) codeword C of an image index S into image blocks. An ECC system [12] in finite field $GF(2^m)$ includes an encoder $Enc_{n,k}(\cdot)$ and a decoder $Dec_{n,k}(\cdot)$, where m is the size of a symbol, n is the number of codeword symbols, and k is the number of message symbols. The code is capable of correcting T_0 or fewer errors. Suppose the image index S is of k symbols, and the number of image blocks is n . Then, let codeword $C = (C_1, C_2, \dots, C_n) = Enc_{n,k}(S)$, where C_i is an m -bit symbol. Additionally, the verifier shares the secret key K with the originator. The scheme includes processes of generating protected images and verifying images. Due to its simplicity and high embedding capacity, LSB watermarking is used in the present scheme although other schemes [13] may be applicable too.

2.1. Generating protected images

Firstly, the originator selects a unique image index S for an embedding image, and then encodes S into codeword $C = (C_1, C_2, \dots, C_n)$. Referring to Figure 2, for each block B_i , let its LSB=0 so as to create a modified block $\hat{B}_i$. Computes the MAC of the image block $\hat{B}_i$ as $h_i = H(\hat{B}_i, i, S, K)$, where $H(\cdot)$ is a one-way function. Sequentially, replaces the LSBs of block B_i with its MAC h_i , block number i , and codeword symbol C_i . Repeats the above process until

all the blocks have been processed. Finally, all the watermarked blocks constitute an authentic image.

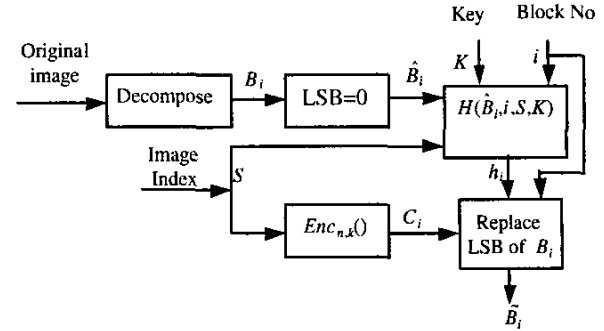


Fig. 2. Embedding image index into LSBs of an image.

2.2. Verifying images

After obtaining a watermarked image, the verifier divides the image into the same blocks as those at the originator side, and extracts index codeword symbol $\tilde{C}_i$ from image block $\tilde{B}_i$. If block $\tilde{B}_i$ is missing, let $\tilde{C}_i = 0$. According to $\hat{S} = Dec_{n,k}(\tilde{C})$, the index is recovered (i.e., $\hat{S} = S$) if the number of tampered blocks is no more than T_0 . The verifier continues to extract the block MAC $\tilde{h}_i$ and the block number i from block $\tilde{B}_i$. Because the verifier knows the secret key K , he calculates $\hat{h}_i = H(\tilde{B}_i, i, \hat{S}, K)$. If $\tilde{h}_i \neq \hat{h}_i$, inspected block $\tilde{B}_i$ is tampered and identified, otherwise, it is authentic. If and only if all the blocks are authentic, the image is trustworthy.

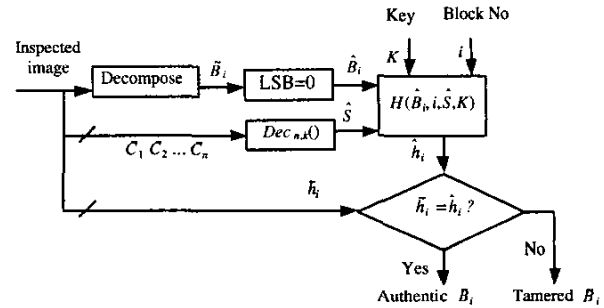


Fig. 3. Detecting tampered blocks.

3. PERFORMANCE

3.1. Security

The security of the localizing scheme solely depends on the underlying hash function. Because the key is secret, half of

the hash bits are enough to provide sufficient security, e.g., a block hash consists of 64 bits if MD5 is employed.

Generally speaking, digital signatures are much longer than MAC for the same security. Therefore, to achieve integrity protection, the present scheme prefers to MAC other than digital signature although digital signature provides integrity protection as well as source authentication. Thanks to the advance in cryptography, a 340-bit Elliptic Curve Digital Signature scheme [14] is roughly the same secure as 1024-bit RSA signature. In other words, the present scheme may be also practical with elliptic curve digital signature.

3.2. Granularity

Table 1 lists the number of bits required for a given security level. It indicates that the present scheme localizes the tampered regions with the finest granularity if $h \geq 64$, where h is the size of MAC.

Table 1. Performance comparison. The block number is represented with 20 bits, and image index is represented with 32 bits. The code for image index is binary in the present scheme. The first column indicates security level.

	Wong[5]	Fridrich[8]	Celik[9]	Present
$O(2^{52})$	104	104	70	73
$O(2^{64})$	116	128	86	83
$O(2^h)$	$h+52$	$2h$	$4h/3$	$h+21$

3.3. Error Resilience

In the proposed scheme, the critical technology is how to recover the image index so as to reconstruct the block hashes. Since Error Correcting Code $Enc_{n,k}$ can tolerate up to T_0 symbol (or block) errors, if there are $t \leq T_0$ tampered blocks, the rest $n - t$ blocks can be verified. To enhance the security, the position for embedding the codeword symbol is determined by the secret key and the block hash so as to increase the attack difficulty.

3.4. Synchronization

It is well known that watermarking is vulnerable to synchronization attack which is hard to defend against. Fortunately, because the attacker's objective is to forge an image other than to invalidate detecting authentic images, the goal of regaining synchronization is to provide a method for detecting the index in case of incidental tampering such as data loss or cropping.

This scheme inserts the block number into the block so as to achieve the synchronization. This robust solution increases the size of block watermark, hence reduces the lo-

calization accuracy. If the block number is removed from the block watermark to improve the localization accuracy without undermining security, there are two methods for regaining synchronization. One is to embed a pattern into the image. For example, the pattern is a random sequence generated from the secret key K . The verifier can recover the pattern so as to synchronize the blocks. Another method is to embed a specified number into an interested block (e.g., center block) and number blocks in the clockwise direction, then, a "sliding-window" is used to regain synchronization.

Accordingly, the block watermark consists of $h+1$ bits if the synchronization technology is applied to improve localizing accuracy.

4. EXPERIMENT

In this example, the 32-bit image index is encoded with ECC to tolerate tampered blocks. The original image shown in Figure 4 is of size 256×256 . It is divided into 1024 blocks. Each block is of size 8×8 and used to embed up to 64 bits, where 11 bits are used for block number, $m = 1$ bit is for the codeword symbol, and 52 bits are for the block MAC to reach the security $O(2^{52})$.

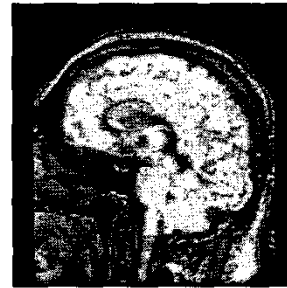


Fig. 4. Original image of size 256×256 .

4.1. Accuracy of localizing tampering

In this simulation, the pixels in rectangle region (80,160; 87,167) is tampered artificially, but other regions are intact. The tampered image is shown as Figure 5. The present scheme localizes the tampered block correctly. The tampered block number is 340 in the scan order.

4.2. Resilience to Cropping

Cropping, one of the simplest image manipulations, cuts the boundary regions but remains a smaller rectangular region of an image. Given an arbitrarily cropped image, the present scheme authenticates unaltered regions. For example, the boundary of the image in Figure 4 is cropped, and the rest

of the image shown in Figure 6 is of size 240×248 . The region (8,8; 247,255) of the original image is authentic.



Fig. 5. Tampered image. The image is regarded to be forged and the tampered block is identified.

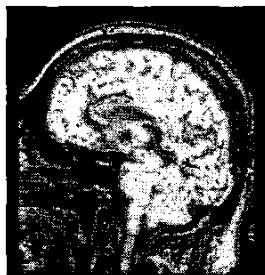


Fig. 6. Cropped image. The boundary regions of the Figure 4 are cut off, but the region (8,8; 247,255) in the original image is verified successfully.

5. CONCLUSION

Many image authentication schemes employ an image index to defeat collage attack. The tree-based scheme [9] does not use the index so as to alleviate management task. However, the tree-based scheme has no advantage over the index-based solutions in computational cost, overhead, localization accuracy, and robustness. Instead of embedding the index into the image directly, the present scheme encodes the image index and inserts the codeword symbols into the image blocks so as to reduce the size of block watermark. Hence, it provides fine localization granularity.

6. REFERENCES

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